

# RAGHAV KACHROO

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Software Engineer drawn to friction points where systems are slow, manual, or breaking under load - building high-throughput data systems and ML infrastructure that fixes them structurally.

## EDUCATION

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| <b>University of California, San Diego</b><br>Master of Science in Data Science (Artificial Intelligence & Machine Learning)      | Sep 2024 – Mar 2026 |
| <b>Indraprastha Institute of Information Technology, Delhi</b><br>Post Graduate Diploma in Data Science & Artificial Intelligence | Sep 2022 – Sep 2023 |
| <b>University of Delhi</b><br>Bachelor of Management Studies                                                                      | Jul 2018 – Jun 2021 |

## EXPERIENCE

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| <b>Amazon</b><br>Software Development Engineer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Jun 2025 – Sep 2025<br>Bellevue, WA    |
| <ul style="list-style-type: none"><li>Built a distributed log indexing and query service over <b>42M+</b> log entries/hour using parallelized binary search, reducing incident triage latency from <b>15+ minutes to under 45 seconds</b>.</li><li>Automated on-call SOPs using AWS Step Functions and Lambda, saving <b>12+ engineer-hours per week</b>.</li><li>Integrated the log query service with internal diagnostic tooling via MCP, adding caching and query batching to maintain <b>sub-2s</b> response times under concurrent incident response load.</li></ul>          |                                        |
| <b>Hao AI Lab, UC San Diego</b><br>Student Researcher                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Jan 2026 – Present<br>San Diego, CA    |
| <ul style="list-style-type: none"><li>Built end-to-end LoRA and full fine-tuning support for Cosmos 2.5 in FastVideo - <b>Cosmos25TrainingPipeline</b> with flow-matching scheduler, 18-channel input, and Reason1 embeddings; <b>2000-step</b> run converged to <b>0.067</b> loss.</li><li>Traced silent training failures across the stack, including LoRA injection failing due to <b>ReplicatedLinear</b> vs <b>nn.Linear</b>, VAE config overwriting by Wan-specific defaults, and hardcoded CFG dropout shape.</li></ul>                                                      |                                        |
| <b>Aark Global</b><br>Software Developer, AI/ML                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Apr 2023 – Sep 2024<br>Delhi, India    |
| <ul style="list-style-type: none"><li>Built an async document ingestion pipeline processing <b>18,000+ pages/day</b> via Azure Queue Storage worker pool, maintaining throughput under variable load.</li><li>Implemented read/write routing during datastore migration - Cosmos DB reads, MongoDB writes - maintaining <b>sub-100ms P95</b> latency throughout cutover.</li><li>Built a full-text search pipeline ingesting scanned PDFs through OCR into indexed Elasticsearch documents, enabling <b>sub-180ms</b> query latency over previously unsearchable content.</li></ul> |                                        |
| <b>Concentrix</b><br>Data Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jun 2022 – Mar 2023<br>Gurugram, India |
| <ul style="list-style-type: none"><li>Replaced sequential scrapers with Airflow-orchestrated Kafka ingestion, increasing throughput by <b>60%</b> and cutting data freshness lag from <b>3 days to 6 hours</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                           |                                        |

## PROJECTS & RESEARCH

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| <b>Flare: ML-Powered Anomaly Detection Pipeline</b>   <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                                                                                      | Jan 2026 – Present  |
| <ul style="list-style-type: none"><li>Built an end-to-end ML anomaly detection pipeline applying Drain3 template mining, Isolation Forest scoring, and DBSCAN clustering to <b>11.2M log lines</b> across <b>575K blocks</b>, achieving <b>0.642 F1</b> at <b>~40ms</b> CPU-only latency.</li><li>Built an LLM-as-judge eval harness grading incident summaries on relevance, specificity, and actionability, scoring <b>4.67/5</b> mean quality without requiring ground truth labels.</li></ul> |                     |
| <b>log-triage-v2: High-Throughput Log Indexing Engine</b>   <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                                                                                | Jan 2026 – Apr 2026 |
| <ul style="list-style-type: none"><li>Go log indexing engine with time-sorted in-memory index, concurrent write ingestion, and binary search reads - sustains <b>42M rows/hour</b> with <b>sub-millisecond</b> nearest-timestamp lookup.</li></ul>                                                                                                                                                                                                                                                |                     |
| <b>TRAI: AI Mobile App to Reduce the Gap Between Triage and Care</b>   <a href="#">IEEE</a>                                                                                                                                                                                                                                                                                                                                                                                                       | Jul 2025            |
| <ul style="list-style-type: none"><li>Deployed a clinical decision support service exposing ML inference via REST APIs, reducing inference time from <b>16s to 5s</b> through caching and request batching.</li></ul>                                                                                                                                                                                                                                                                             |                     |

## TECHNICAL SKILLS

**Languages:** Python, Go, SQL, Java

**Backend & Infra:** FastAPI, REST APIs, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, MCP

**Distributed & Data Systems:** Kafka, Spark, Airflow, Elasticsearch, MongoDB, Cosmos DB, Azure Queue Storage, Redis

**Observability & Cloud:** Prometheus, OpenTelemetry, AWS (Lambda, Step Functions, S3), Azure

**AI & ML:** PyTorch, Scikit-Learn, Hugging Face Transformers, LangChain, MLflow, RAG Pipelines, Diffusers, LoRA

**Concepts:** Distributed Systems, Concurrency, MLOps, Inference Optimization, System Design, Model Monitoring